

The background features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

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CON317

Amazon ECS networking deep dive

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Agenda

- 01 Amazon ECS
- 02 What users need
- 03 Networking modes in Amazon ECS
- 04 Connecting to other AWS services
- 05 Interconnecting Amazon ECS services
- 06 Anycast instead of DNS
- 07 Call to action
- 08 Q&A

About your speakers



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Please ask us questions



Amazon ECS: Significant scale



Over 2.25 billion

tasks launched every week

Tens of thousands

of API requests served per second

32 AWS Regions, 6 continents

Global presence

**65% of all new AWS
containers customers**

use Amazon ECS

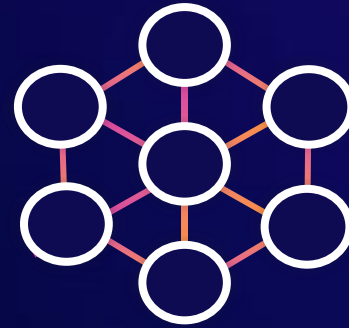
What users need on Amazon ECS



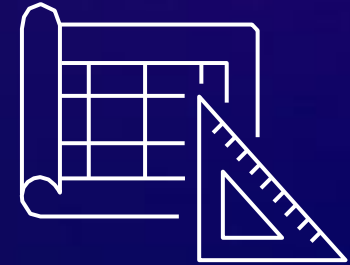
Foundations for
networking



Connections to
AWS services



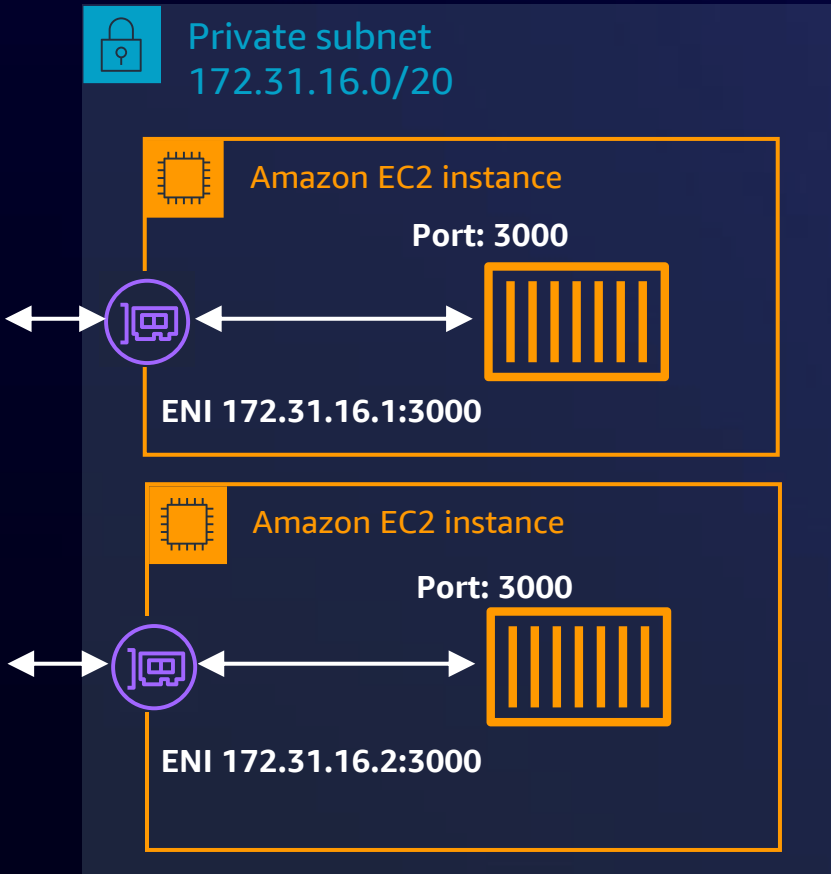
Interconnecting
Amazon ECS
services



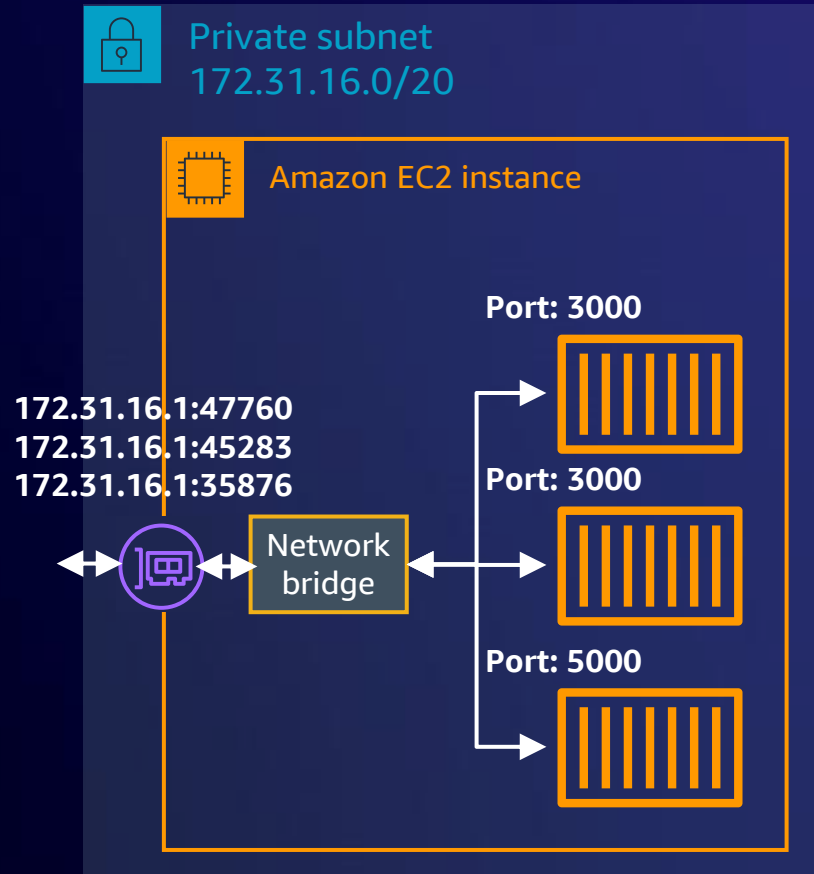
Networking best
practices

Amazon ECS: Choosing a network mode

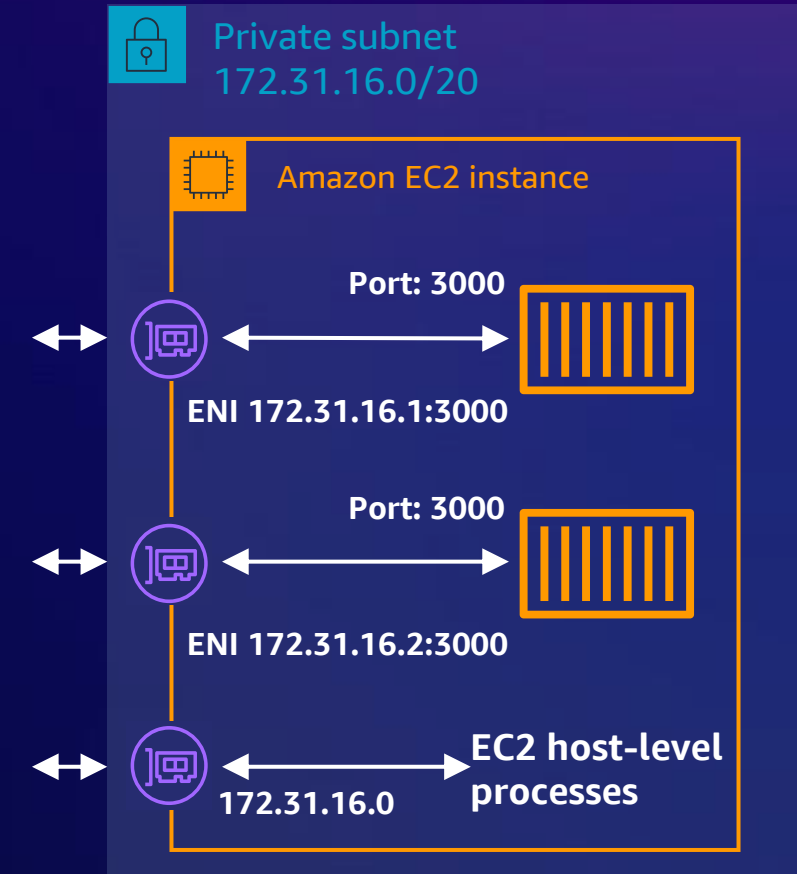
host mode



bridge mode



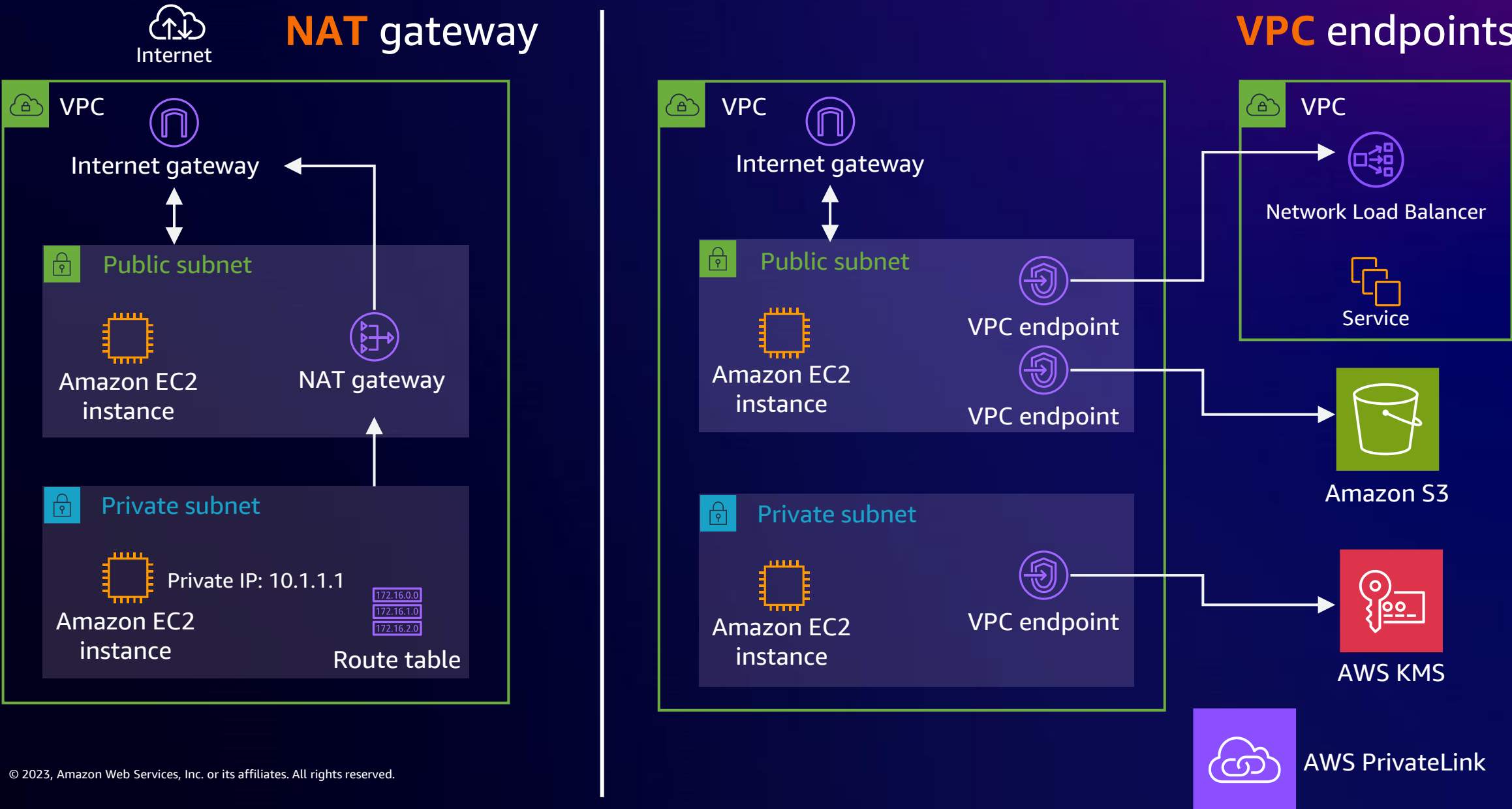
awsvpc mode



Amazon ECS and AWS Fargate: Networking modes

Mode	 Amazon ECS	 AWS Fargate
Bridge	YES	NO
Host	YES	NO
awsvpc	YES	YES
None	YES	NO

Outbound and AWS service connectivity



Interconnect your Amazon ECS services

Amazon ECS service discovery

Elastic Load Balancing (ELB)

Amazon ECS Service Connect

AWS App Mesh



Using Amazon ECS service discovery

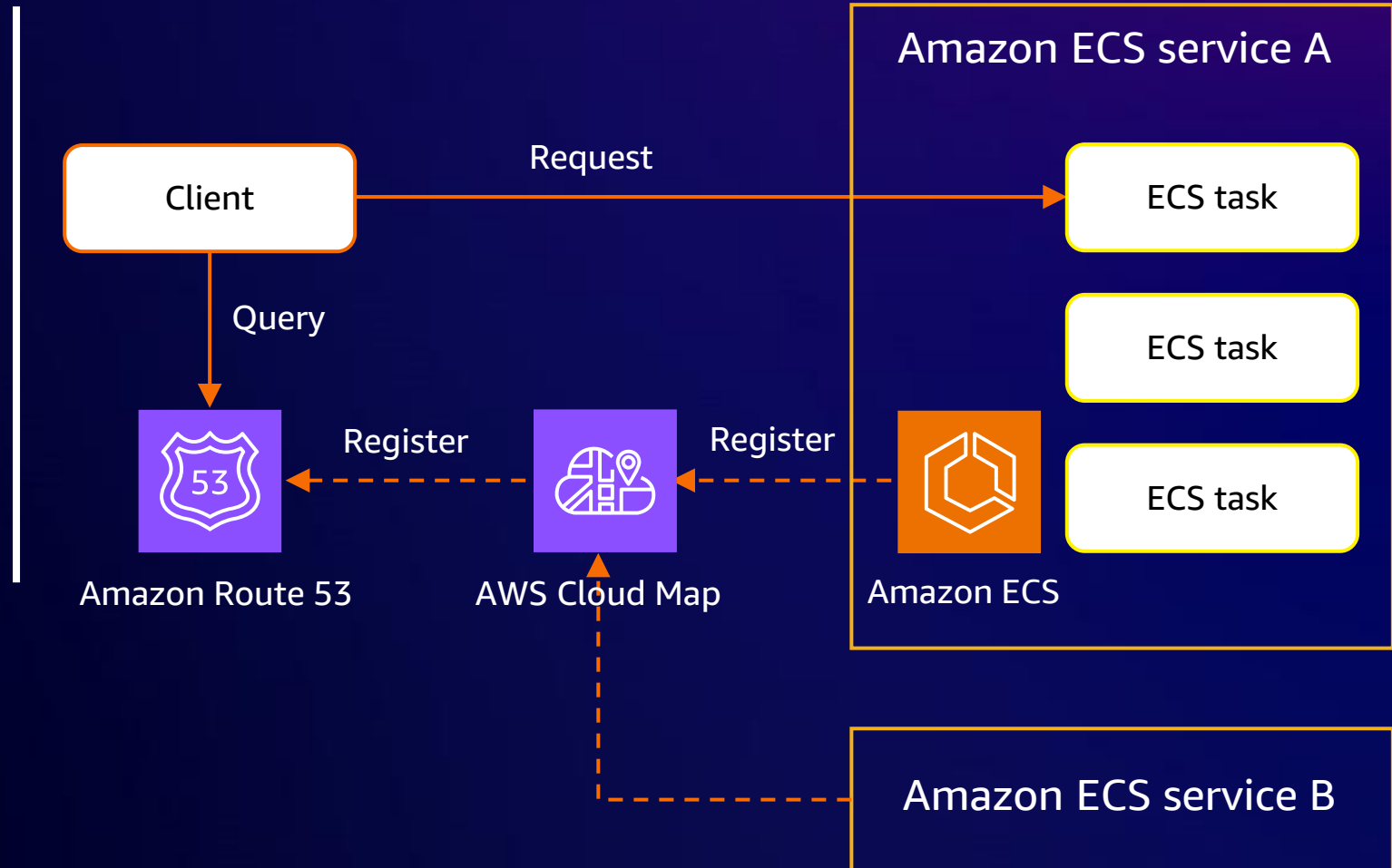
Simple DNS discovery

Clients connect directly to providers

Fewer components in the system

No traffic telemetry

DNS offers only basic discovery



Using Elastic Load Balancing (ELB)

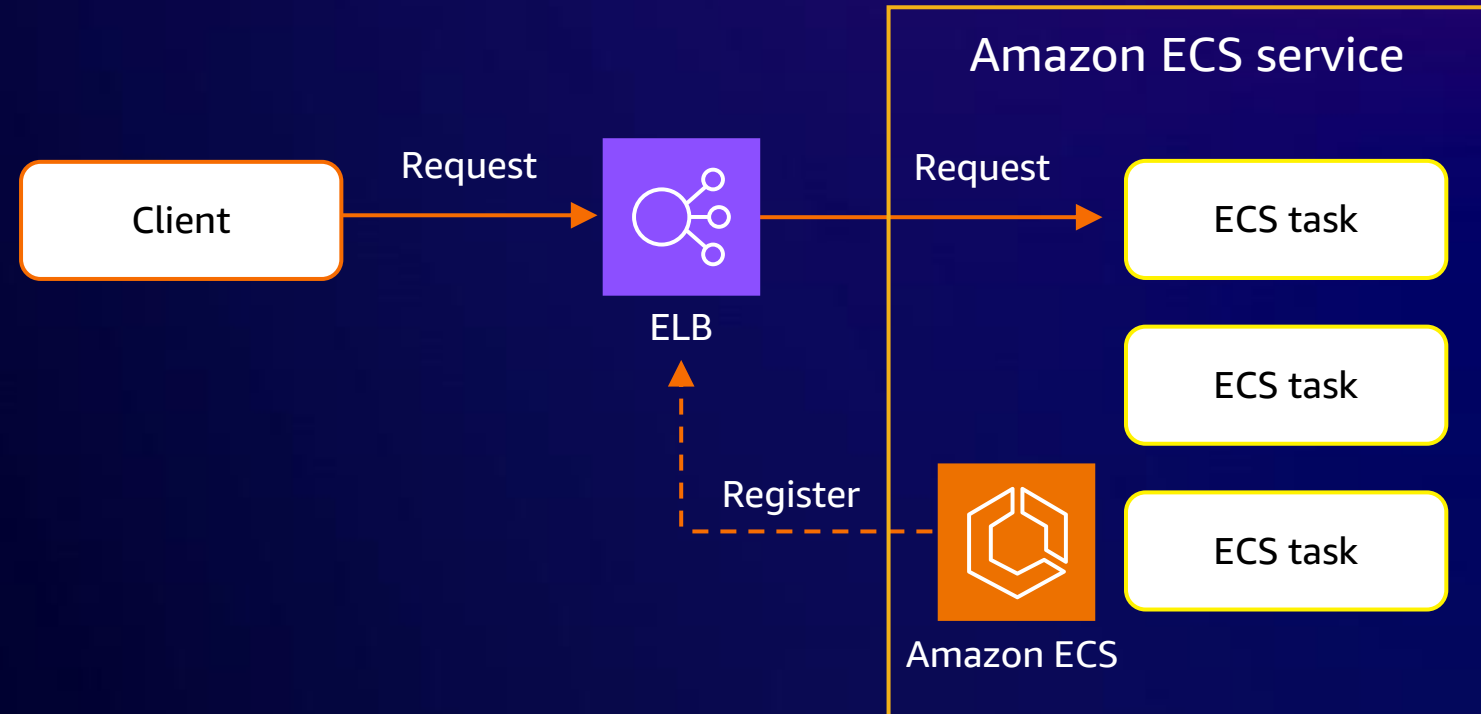
Discovery is abstracted away

Rich feature set

Traffic telemetry

Additional infrastructure

Additional latency



Using AWS App Mesh

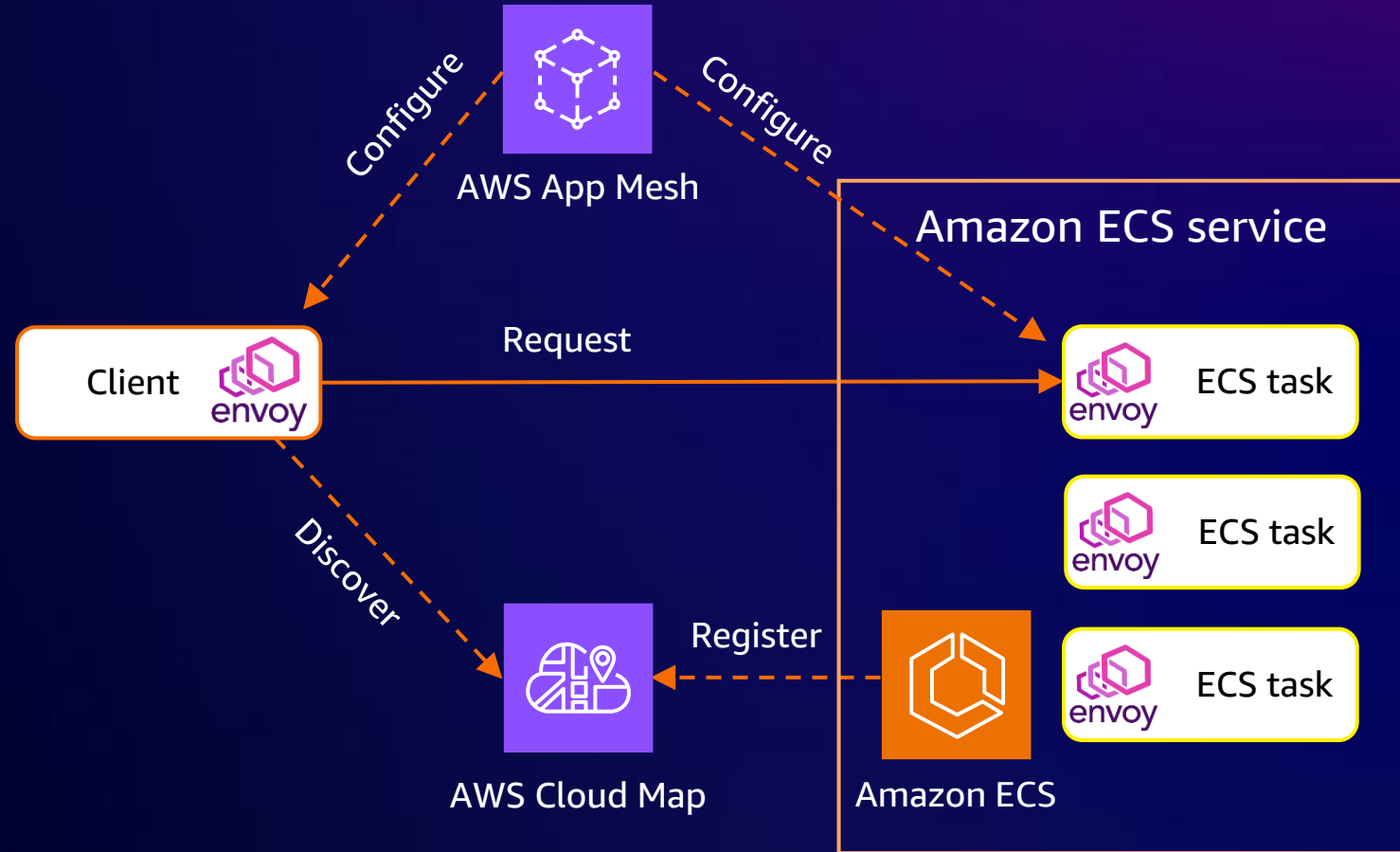
Rich traffic observability

Fine-grained traffic controls

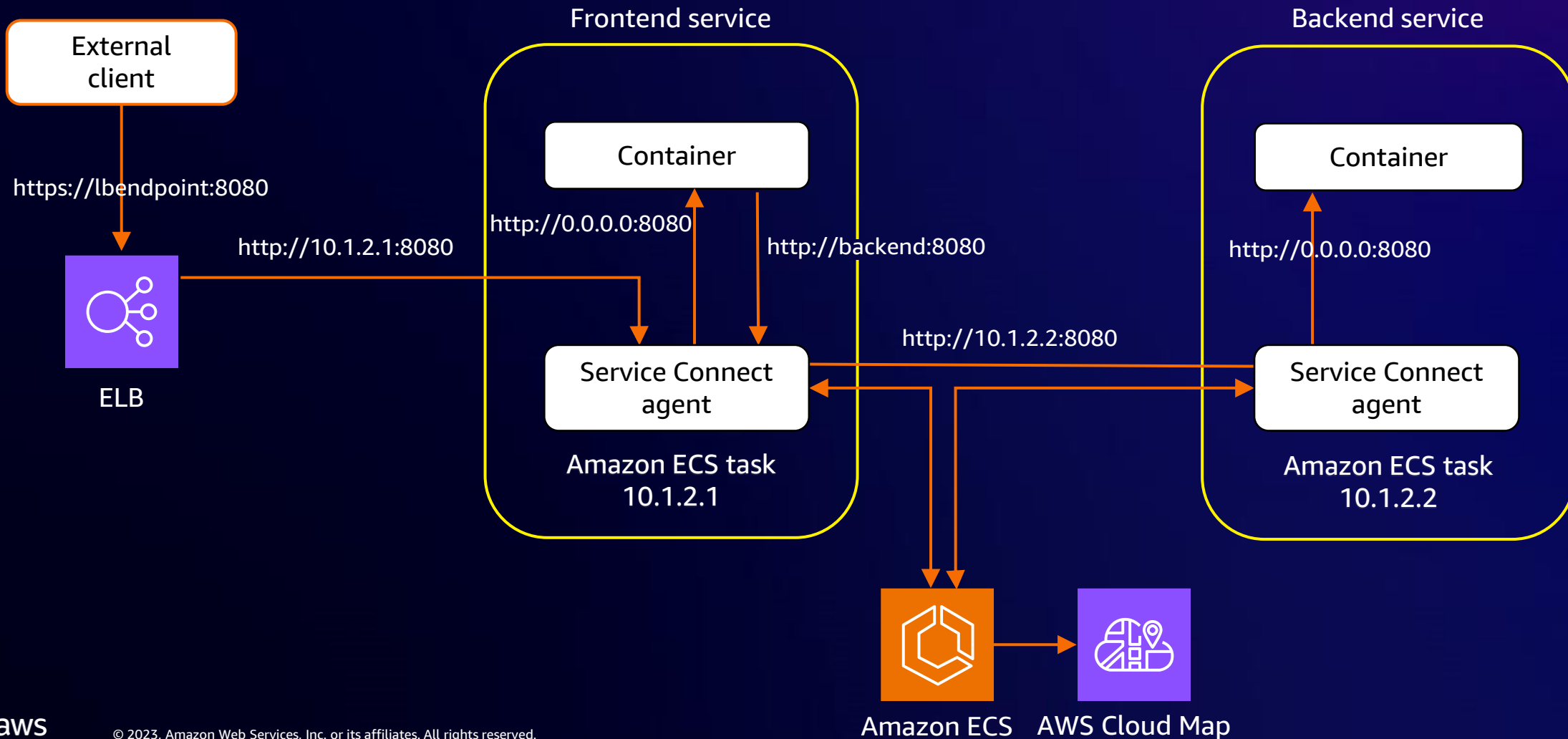
Encryption and authentication

Additional components to manage

Flexibility comes with complexity

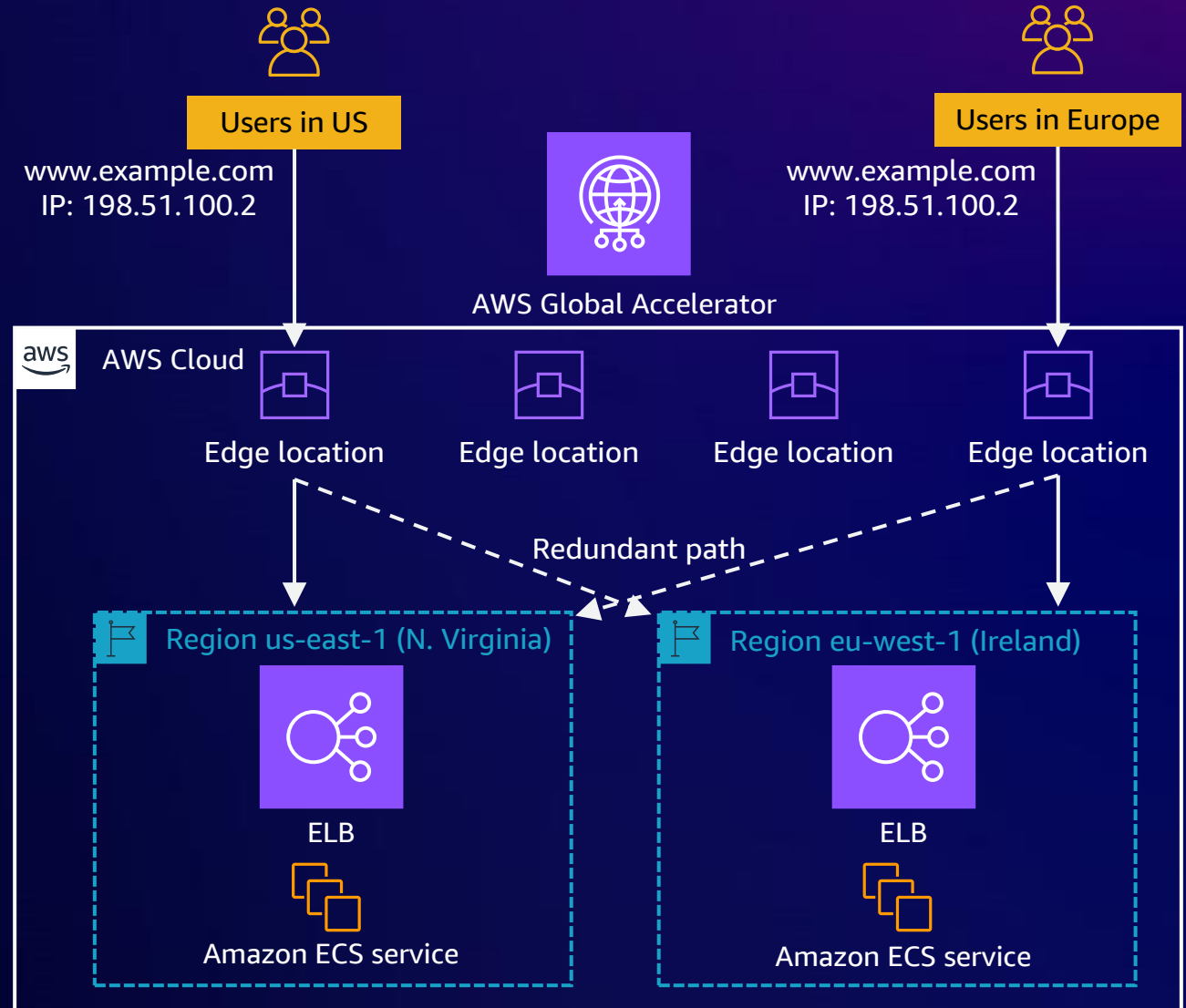


Amazon ECS Service Connect



Anycast instead of DNS: AWS Global Accelerator

- Uses AWS global network from edge to Region
- Client traffic ingresses via closest available edge location
- Route client to closest healthy endpoint
- No DNS switchover required, same IP address globally (static IP anycast)



Call to action



**Amazon ECS
workshop**



**Amazon ECS
best practices**



**Amazon ECS
Service Connect
workshop**

Thank you!



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